MOTION IN ONE DIMENSION

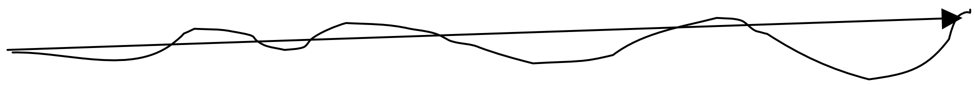
DEFINITIONS

DISPLACEMENT (s): change of position of a body in a given direction.
i.e. a vector

DISTANCE: how far a body moves in any direction.
i.e. a scalar

EXAMPLE

A car moves along West Coast Drive from Sorrento to Trigg. Its motion is represented in the diagram below.



The distance on the road is 5.30 km. (The direction is not important and the distance is the sum total of all movements.)

The displacement (measured from the start of the motion to the finish) is 4.75 km S 5.0° E.

VELOCITY (v): the rate of change of position in a given direction.
i.e. a vector

$$\boxed{\mathbf{v = \frac{s}{t}}}$$

where v = the average velocity for the motion (ms^{-1})
s = the displacement of the object or body (m)
t = the time taken for the entire motion (s)

SPEED: the rate of change of distance.
i.e. a scalar

$$\boxed{\mathbf{speed = \frac{d}{t}}}$$

where d = the total distance travelled by the object (m)
t = time taken for the entire movement (s)

EXAMPLE

A competitor in a rogaining competition covers 2.50×10^2 m south in 65.0 s, 3.50×10^2 m west in 1.20×10^2 s and 4.00×10^2 m north in 1.05×10^2 s. Calculate:

(a) the distance covered in this time.
(b) the displacement of the competitor.
(c) the average speed maintained.
(d) the average velocity of the competitor.

ACCELERATION (a): the rate of change of velocity in a given direction.
i.e. a vector

$$\boxed{\mathbf{a = \frac{\Delta v}{t} = \frac{v - u}{t}}}$$

where a = the acceleration of the object (ms^{-2})
v = final velocity of the object (ms^{-1})
u = initial velocity of the object (ms^{-1})
t = time taken for the acceleration to occur (s)

For uniformly accelerated motion, the following equation can be used where appropriate.

$$\boxed{v_{ave} = \frac{u + v}{2}}$$

where v_{ave} = the average velocity for the accelerated motion (ms^{-1})

EXAMPLE
A car moving at 60.0 kmh^{-1} along a level road brakes sharply to avoid a ball that has rolled down a driveway. It decreases uniformly to 20.0 kmh^{-1} in 2.60 s. Calculate the acceleration experienced by the car.

EQUATIONS OF MOTION

For objects and bodies undergoing uniform acceleration, a series of equations can be derived to describe the motion.

$$\mathbf{v = u + at}$$

$$\mathbf{s = ut + \tfrac{1}{2}at^2}$$

$$\mathbf{v^2 = u^2 + 2as}$$

where
v = final velocity (ms^{-1})
u = initial velocity (ms^{-1})
a = aceleration (ms^{-2})
t = time taken for the acceleration (s)
s = displacement during the acceleration (ms^{-2})

These equations can be used to solve problems involving objects moving horizontally across the ground, down inclined planes such as hills, or vertically under the influence of Earth's gravity.

Near the Earth's surface, the acceleration due to gravity is accepted to be:

$$\mathbf{g = 9.80\ ms^{-2}}.$$

EXAMPLE 1
A car moving at 60.0 kmh^{-1} along a level road brakes sharply to 20.0 kmh^{-1} in 2.70 s in order to miss a ball that had rolled down a driveway. Calculate the acceleration experienced by the car.

EXAMPLE 2
A boy playing catch by himself throws a ball vertically upwards at 12.0 ms^{-1} and catches it at the same height as the release point. Calculate:

(a) the maximum height reached above the release point.
(b) the time of flight for the ball.

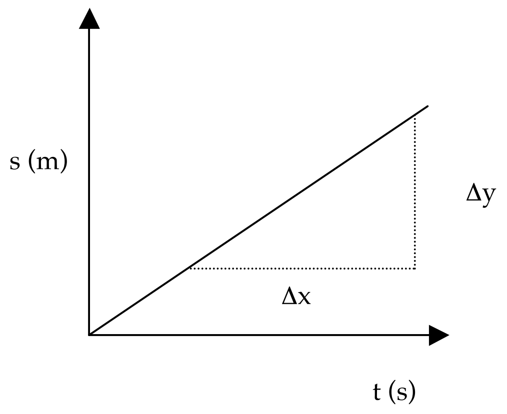
EXAMPLE 3
A girl playing with a racquet and ball hits the ball vertically upwards at 25.0 ms^{-1}. It rises above a tree and gets caught by a branch on the way down. It finishes 5.00 m above the point at which it was struck.

(a) Calculate the time of flight for the ball.
(b) Determine the velocity of the ball as it hits the branch.

MOTION GRAPHS

DISPLACEMENT - TIME

CONSTANT VELOCITY

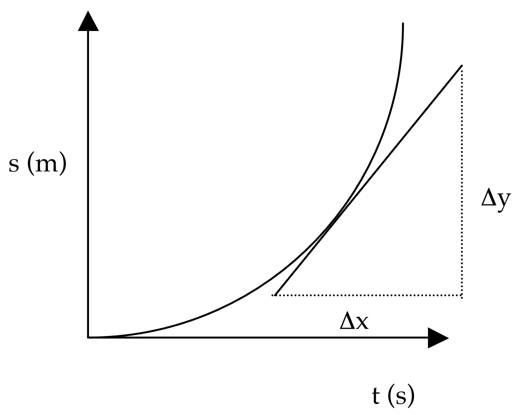


The slope or gradient of the line is given by:

$$\text{slope} = \frac{\Delta x}{\Delta y} = \frac{s}{t}$$

Hence the slope of the graph gives the **constant velocity** of the body or object.

CONSTANT ACCELERATION



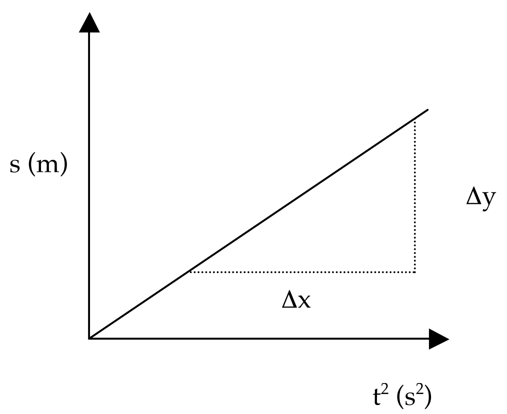
The graph at left represents an object falling under gravity or one that is uniformly accelerated across the ground.

The slope at a point on the curve gives the **instantaneous velocity** of the object at that time.

The graph is a parabola and clearly suggests that $s \propto t^2$.

The equation relating displacement and acceleration is $s = ut + 1/2\ at^2$.

If $s \propto t^2$, then plotting s versus t^2 should produce a straight line graph.



The slope or gradient represents 1/2 a.

The intercept represents ut.

The general equation of a line is:

$$y = mx + b$$

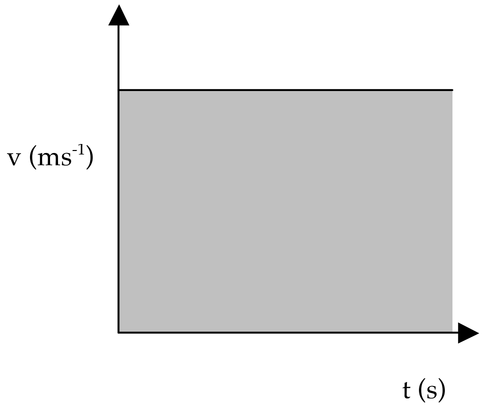
b represents ut while m = 1/2 a.

IMPORTANT

The slope of a linear graph generally means something - it is related to an equation that is applicable. You must know what the slope or gradient represents!

VELOCITY - TIME

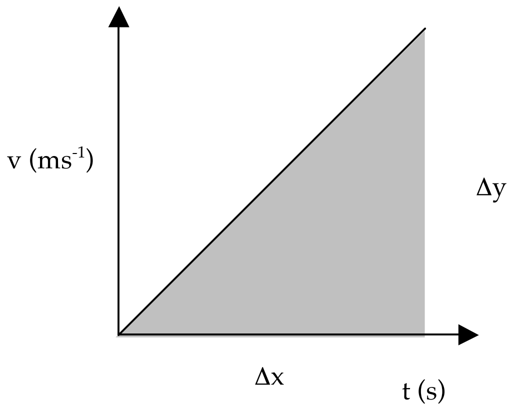
CONSTANT VELOCITY



The area under a v - t graph represents the displacement for the body.

i.e. $s = v \times t$

CONSTANT ACCELERATION



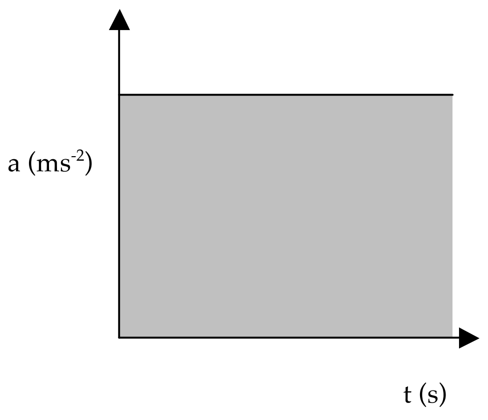
The slope gives the **acceleration** of the body.

$$\text{slope} = \frac{\Delta v}{\Delta t} = a$$

Area under graph = displacement
= 1/2 x base x height

ACCELERATION - TIME

CONSTANT ACCELERATION



Area under graph = change in velocity
= Δv

EXAMPLE

A car travels at 20.0 ms^{-1} as it passes an observer on the side of the road. The car maintains this velocity for 4.00 s before decelerating uniformly for 3.00 s to 10.0 ms^{-1}. In the next 4.00 s it uniformly increases to 25.0 ms^{-1} before continuing at constant velocity. The entire motion occurs in a straight line.

(a) Draw a velocity - time graph for the first 11.0 s of the motion.
(b) Calculate the displacement of the car in this time.
(c) Draw an acceleration - time graph for the motion.